



JUNGHUN CHAE

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EDUCATION

M.S.–Ph.D. Physics Ulsan National Institute of Science and Technology (UNIST), Korea	Mar. 2021 – present
Bachelor of Science Physics & Computer Science (Minor) Ulsan National Institute of Science and Technology (UNIST), Korea	Mar. 2017 – Feb. 2021

PUBLICATIONS

1. **Chae, J.[†]**, Kim, W.[†], Jung, W.[†], Jeong, D.[†], Lim, R.[†], Chamlagain, M., Jung, G., et al. *Warning signals for the long-term innovative potential of genetic research based on 20 million papers and patents. Under Review*
2. **Chae, J.[†]**, Lim, R.[†], Martin, T. L. P.[†], Ghim, C.-M., and Kim, P.-J. *Enlightening the blind spot of the Michaelis–Menten rate law: The role of relaxation dynamics in molecular complex formation*, *Journal of Theoretical Biology* **597**, 111989 (2025).
3. Lim, R.[†], Martin, T. L. P.[†], **Chae, J.[†]**, Kim, W., Ghim, C.-M., and Kim, P.-J. *Generalized Michaelis–Menten Rate Law with time-varying molecular concentrations*, *PLOS Computational Biology*, **19**(12), e1011711 (2023).
4. **Chae, J.[†]**, Lim, R.[†], Ghim, C.-M., and Kim, P.-J., *Backward simulation for inferring hidden biomolecular kinetic profiles*, *STAR Protocols* **2**(4), 100958 (2021).
5. Lim, R.[†], **Chae, J.[†]**, Somers, D.E., Ghim, C.-M., and Kim, P.-J., *Cost-effective circadian mechanism: rhythmic degradation of circadian proteins spontaneously emerges without rhythmic post-translational regulation*, *iScience* **24**, 102726 (2021).

- [†] indicates the co-first authorship
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- Google Scholar: <https://scholar.google.com/citations?user=r1rxOA0AAAAJ>

WORK EXPERIENCE

High-profile reconstruction of metabolic network of <i>Segatella Copri</i> Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2025–
Game thoery in Vicsek model Advisor: Cheol-Min Ghim (UNIST)	2025
Information processing in biological systems Advisor: Cheol-Min Ghim (UNIST)	2024
Gene bibliometric review on papers and patents Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2023–24
Michaelis-Menten kinetics for circadian proteins Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2022–24
Dynamic flux balance analysis on mammalian gut microbiome Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2021
Rhythmic degradation of circadian proteins Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2020

TECHNICAL EXPERIENCE

Multi-agent simulation with GPU (CUDA) Advisor: Cheol-Min Ghim (UNIST)	2025
Big Data Analysis using PubMed & PATSTAT Data Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2024
Large Matrix Diagonalization ($\geq 10,000$) with high precision Advisor: Cheol-Min Ghim (UNIST)	2024
Reinforcement Learning & Deep Learning Advisor: Cheol-Min Ghim (UNIST) & Hyunmo Yang (UNIST)	2022
Gillespie algorithm & Linear Noise Approximation Advisor: Cheol-Min Ghim (UNIST)	2022
LP (Linear Programming) Problem Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2021
Numerical ODE Problem Advisor: Cheol-Min Ghim (UNIST) & Pan-Jun Kim (HKBU)	2020

SKILLS

- **Languages:** English (Proficient), Japanese (Proficient), Korean (Native)
- **Programming:** Python (NumPy, SciPy, Matplotlib, Pandas, FEniCS, ...), C/C++ (CUDA C++, libTorch, ...)
- **Software:** Linux, Git, Inkscape, ...
- **Bio-Software:** Metaphlan, Kraken, QIIME, ...
- **Miscellaneous:** $\text{\LaTeX}$, Nsight System

CONFERENCES AND PRESENTATIONS

Korean Physical Society Game Theory in Mobile Agents - Introduction of multilevel selection	Oct. 2025
The 23rd Workshop for Statistical Physics Game Theory in Mobile Agents - How initial configuration changes dynamics	Aug. 2025
Korean Physical Society Microbiome Dynamics under Dietary Shifts	Apr. 2025
Korean Physical Society A Scientometric Study on the Innovative Frontiers in Genetic Research	Oct. 2024
Korean Physical Society Improving the blind spot of the Michaelis–Menten rate law	Apr. 2024
International Conference on Biological Physics (ICBP 2023, Poster Session) Generalized Michaelis-Menten Rate Law with Time-varying Molecular Concentrations	Aug. 2023
Korean Physical Society Effective rate law for biological feedback control	Apr. 2023
Winter Q-Bio (Poster Session) Michaelis–Menten Rate Law beyond Adiabatic Approximations	Feb. 2023
International Conference on Systems Biology (ICSB 2022) Rhythmic protein degradation for cost-effective circadian oscillation	Oct. 2022
Korean Physical Society The Michaelis-Menten Rate Law beyond the Steady State and Its Application to Circadian Dynamics	Apr. 2022
Biophysical Society Annual Meetings (Poster Session) Post-translational modification for cost-efficient circadian rhythms	Feb. 2022

TEACHING EXPERIENCE

TA: Network Science Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea	Fall 2025
TA: Classical Mechanics II Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea	Fall 2022
TA: Classical Mechanics I Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea	Spring 2022
TA: Network Science for Complex Systems Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea	Fall 2021
TA: History of Science and Technology Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea	Spring 2019

OTHER EXPERIENCES

Peer review Scientific Reports	Winter 2021
Curriculum development Network Science and Machine Intelligence (Dept. of Physics, UNIST)	Fall 2021
Research Assistant Hongkong Baptist University (April to Sept. 2021)	Summer 2021